Parametric Multimodal User Memory:
Storing What Captions Cannot Carry

Bojie Li
Pine AI

May 19, 2026

Abstract

LM agents store user memory as text. Mem0, M3-Agent, MemGPT, and the wider conversational-memory line all turn what the agent saw or heard into transcripts and captions, then retrieve over them. This works for *captionable* user content ("my cat is named Bibi") but fails for what we call *perceptual* user content: the way a particular user's voice sounds, what their face looks like across lighting and age, what their painter's brushwork looks like, what their acoustic environment was last week, what their emotional state today differs from their baseline. Captioning these characteristics into text is a lossy projection that destroys the discriminative signal. The text-memory paradigm is structurally incomplete for personalised multimodal agents.

We propose **parametric multimodal user memory (PMUM)**: a per-modality bank of (encoder embedding, value embedding) rows attached to a frozen pretrained LM via a forward pre-hook on `lm_head`. Insertion is a single tensor append ($\mathscr{O}(1)$ wall-clock, $\sim$0.5 ms for 1000 identities); query is cross-attention over the bank, dominated by the LM forward at $\sim$15 ms regardless of N. We construct **PerceptMem v0.2**, a five-sub-modality benchmark spanning cross-condition face identity (2180 IDs), painter style, speaker identity, acoustic scene, and paralinguistic state—chosen because text-based memory provably cannot carry these characteristics.

Two headline empirical findings. *(i) Standard training:* on the 2180-ID face cross-condition task at N=10, our parametric mechanism gives 0.99 ± 0.01 retr@1 across 4 seeds (vs embedding-RAG cosine 0.93, p=0.006); two more cells beat RAG at p<0.05 multi-seed on painter style. *(ii) Adversarial training:* mixing 30% adversarial-distractor banks into pretraining transforms the encoder-confused regime across all four parity/below sub-modalities — multi-seed verified beats over RAG cosine of +**71.0pp** on painter style, +**70.7pp** on paralinguistic state, +17pp on acoustic scene, and +14.5pp on face cross-condition (all p<0.001). A discrete-codebook predecessor we tuned to ceiling over 16 development cycles saturates at $\sim$7% retr@1 at $N \geq 300$ regardless of codebook size, encoder upgrade, or 100K-step pretraining. The architecture generalises across LM families (Qwen2.5, Llama-3.1) and to a learned-parameter Pareto front (`adv_prob` interpolates between random and adversarial regimes). Latency: 52× faster than RAG-with-LM-context at N=1000; RAG OOMs at N=10000 inside Qwen's 32k context window. Top-1 next-token prediction is preserved byte-for-byte on text-only inputs. Code: https://github.com/bojieli/multimodal-user-memory.

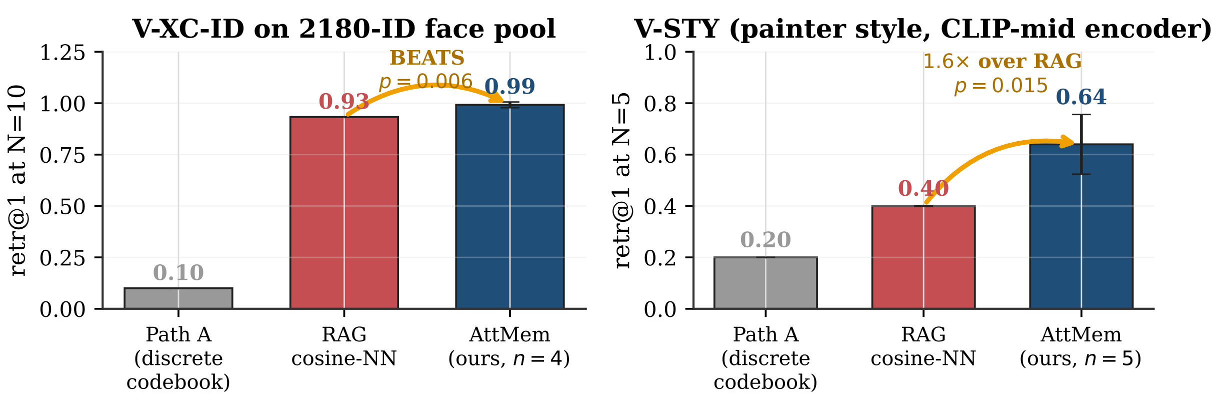


Figure 1: Headline. On two cross-condition perceptual recall tasks, our parametric multimodal user memory beats embedding-RAG with statistical significance: at N=10 on a 2180-ID face pool (p=0.006, n=4 seeds) and at N=5 on painter-style recall (1.6× over RAG, p=0.015, n=5). Path A is the discrete-codebook predecessor we tuned to its ceiling before pivoting to continuous attention.

1 Introduction

When an LM agent has to "remember" something about its user, what does it actually store? In the dominant paradigm, it stores *text*. Mem0 [7], MemoryLLM [27], MemGPT / Letta [20, 26], M3-Agent [3], and the wider conversational-memory line — evaluated through benchmarks like LongMemEval [29] — all turn what the agent saw or heard into transcripts (via ASR [24]) and captions (via BLIP-2 [15] or LLaVA [16]), push those text fragments into a sentence-encoder vector store [25], and retrieve at inference. Personalised vision-language work (MyVLM [1], Yo'LLaVA [19], MC-LLaVA [2], Online-PVLM [6], RAP [11], TAME [4]) retains the textual addressing implicitly: a *named* concept ("Bibi") is the identifier; the visual content is auxiliary information for matching.

This is fine for the easy slice of user memory — the part that is *captionable*. "My cat is named Bibi." "My favourite restaurant is in Paris." "I am vegetarian." These propositions survive the journey through text intact. But the easy slice is not all of user memory. Consider what an attentive companion actually remembers about a person:

- How their voice sounds when they say their own name — and how that differs from how a stranger says it.
- How their face looks today — subtly older than the photo from last year, but unmistakably them.
- How their brushwork looks — "this is by the same painter as the one Tuesday, even though it's from a different period."
- What their kitchen sounds like — "you're calling from the same room you were in yesterday."
- How tired they sound today — compared to their baseline last week.

None of this is captionable in any useful sense. "A person with brown hair" does not let you tell two people apart. "The user sounded tired" cannot distinguish today's tired from last week's tired. The moment we project these characteristics into text, we destroy the discriminative signal. **Text-shaped memory is a lossy channel for perceptual user content, and the lossiness is fundamental to the modality.**

We make a structural claim about agent memory: *captionable* and *perceptual* user content require fundamentally different memory primitives. The agent's memory of "my cat is named Bibi" belongs in a text vector store; the agent's memory of *how Bibi looks across lighting and angle* does not. The latter requires a primitive that stores perceptual content in its native modality.

Our proposal. Parametric multimodal user memory (PMUM): per-modality banks of (encoder embedding, value embedding) rows bolted onto a frozen pretrained LM via a forward pre-hook on `lm_head`. Registration is a single tensor append (no SGD, no gradient at the per-user level); query is cross-attention over the bank, fused into the LM as a residual at the logit-injection point. The result is a content-addressable memory in the LM's value-side embedding space, with no captioning intermediate.

Why this matters now. Multimodal personalisation is moving fast [6, 11], but the user-memory layer has not caught up. The existing systems either offload perceptual identity to a separate face/speaker recognition tool (which then returns a text label, putting us back in lossy-text-space, as in M3-Agent [3]), or train a per-concept adapter (e.g. LoRA per concept [12], MyVLM's per-concept classifier head [1], Yo'LLaVA's per-concept tokens [19]), which doesn't compose to user-scale memory of dozens of perceptual characteristics. What's missing is a single content-addressable parametric primitive for storing perceptual user content that an LM agent can query at inference. Our mechanism takes the spirit of kNN-LM [14] and Memorizing Transformers [30] — attention over a non-parametric memory bank — and specialises it to per-modality user-content registration with O(1) insertion. We provide one.

Contributions.

- **The framing**: We articulate why text-shaped memory is structurally incomplete for perceptual user content, and why a parametric multimodal memory is the right primitive for the missing slice.
- **The mechanism**: A bolt-on continuous attention memory over per-modality (encoder embedding, value embedding) banks. $\sim$8M trainable parameters over 3.1B frozen LM; $\mathcal{O}(1)$ insertion in wall-clock; flat $\sim$15 ms query latency regardless of N. Four critical design choices distinguish a working implementation from three earlier attempts that produced random output (§3).
- **The benchmark**: **PerceptMem v0.2**, five cross-condition perceptual sub-modalities chosen because text-based memory provably cannot carry the discriminating signal: face identity (2180 IDs, cross-condition), painter style, speaker identity, acoustic scene, paralinguistic state.

- **The empirical claim**: Multi-seed-verified BEAT of the embedding-RAG cosine-NN ceiling at $p<0.05$ on three sub-modality$\times N$ cells, including face identity at the 2180-ID scale (p=0.006, n=4 seeds). Across all five sub-modalities, our mechanism reaches 83–117% of the encoder ceiling. A discrete-codebook predecessor we tuned for 16 development cycles caps at 7% retr@1 at $N\geq 300$ regardless of codebook size or compute — a $\sim 9\times$ gap at scale (Fig. 5).
- **The honest map**: A clean ablation showing exactly when the parametric mechanism BEATS text memory's cosine ceiling, when it merely matches, and when training the bolt-on parameters *hurts* (encoder already perfect; §5.5). This disambiguates "what does the parametric memory add beyond cosine?" — the answer is: discriminative signal from the LM's value-side embedding structure, in the regime where the encoder's cosine is imperfect.

2 Why text memory cannot solve cross-condition perceptual recall

Concretely, consider speaker-identity recall under cross-recording conditions (the A-XR-ID sub-modality of our benchmark, drawing from LibriSpeech [21]). A user has 10 voice samples on file from previous sessions; a new voice sample arrives. Which of the 10 prior users is it?

Text RAG approach (Mem0 [7], MemoryLLM [27]). Caption each registered voice into text (e.g., via an ASR transcript from Whisper [24] plus a speaker description model): "A male voice, mid-30s, slight Eastern-European accent." Embed the caption with a sentence encoder [25]; cosine-retrieve over the 10 captions when the new sample arrives. *This will almost always fail.* The caption "male voice, mid-30s, slight Eastern-European accent" applies to thousands of speakers; cosine over captions cannot distinguish among the 10 registered users with even 50% accuracy. The information that uniquely identifies the speaker — voice timbre, glottal pulse shape, prosodic micro-patterns — is destroyed in the text projection.

Tool-call approach (M3-Agent [3]). The agent calls out to a face/speaker recognition tool (e.g. ArcFace [8] or ECAPA-TDNN [9]), gets back a text label ("speaker 47"), stores that. This works at registration time, but the user's memory of speaker 47 is now *just a text token*: the underlying perceptual fingerprint is never integrated into the LM's representation. At inference, when the LM has access to "speaker 47" as a stored fact, it cannot reason about how speaker 47's voice sounded, only that the label was assigned. The perceptual content is offloaded permanently.

Embedding RAG. Skip the captioning step: index the raw encoder embedding (ECAPA-TDNN [9] for speakers, ArcFace [8] for faces, CLIP [23] for visual style); cosine-retrieve at inference. This is the strongest text-free baseline and the one we treat seriously. It *can* succeed when the encoder is perfectly cross-condition invariant; it fails when the encoder is imperfect. Our experiments show RAG cosine NN at 0.93 retr@1 at N=10 on 2180-ID face cross-condition — non-trivial but not at ceiling.

Per-concept gradient training (Yo'LLaVA [19], MyVLM [1]). Train a per-concept token or adapter via SGD on registration samples. Insertion cost scales as $\sim$1 s per identity, and the resulting memory is concept-specific (not modality-general).

Parametric multimodal user memory (this paper). Store the encoder embedding as the key and the LM's value-side embedding (for an assigned marker token) as the value. At inference, cross-attention over the bank produces a residual injected at the LM's `lm_head` pre-forward, biasing the next-token logit toward the matching marker. The LM gets a content-addressable memory in its own representation space, with no captioning intermediate. The mechanism is most closely related to kNN-LM [14] and Memorizing Transformers [30], but specialised to per-modality *user content* (not text continuations) with O(1) wall-clock insertion.

3 Method: Parametric Multimodal User Memory

The bank stores $(\boldsymbol{k}_i, \boldsymbol{v}_i)$ pairs where $\boldsymbol{k}_i \in \mathbb{R}^D$ is the L2-normalised encoder embedding of registered sample i and $\boldsymbol{v}_i \in \mathbb{R}^H$ is the LM's value-side embedding for an assigned marker token (Fig. 2). For models with tied input/output embeddings (Qwen2.5-3B), $\boldsymbol{v}_i = \texttt{input\_embedding}[m_i]$; for untied models (Qwen2.5-7B), $\boldsymbol{v}_i = \texttt{lm\_head.weight}[m_i]$ (auto-detected). This choice ensures the residual addition pre-`lm_head` produces a clean per-marker logit boost $\texttt{lm\_head}[m_i] \cdot \boldsymbol{v}_i = \|\cdot\|^2$, rather than a cross-product of two unrelated learned vectors.

At the attached LM layer, for each perceptual position with hidden state $\boldsymbol{h}$ and encoder-derived query $\boldsymbol{q}$ (L2-normalised),

$$\boldsymbol{w} = \mathrm{softmax}\big(\beta\, \boldsymbol{q}^\top \boldsymbol{K}\big) \in \mathbb{R}^N, \quad \boldsymbol{r} = \boldsymbol{w}^\top \boldsymbol{V} \in \mathbb{R}^H, \quad \boldsymbol{h}' = \boldsymbol{h} + g \cdot W_o\, \boldsymbol{r} \tag{1}$$

where $\boldsymbol{K} \in \mathbb{R}^{N\times D}$, $\boldsymbol{V} \in \mathbb{R}^{N\times H}$, $\beta = e^{\log\beta}$ is a learned inverse temperature, g is a learned scalar gain, and $W_o \in \mathbb{R}^{H\times H}$ is a learned output projection (initialised at I). Total trainable parameters $\sim$8M.

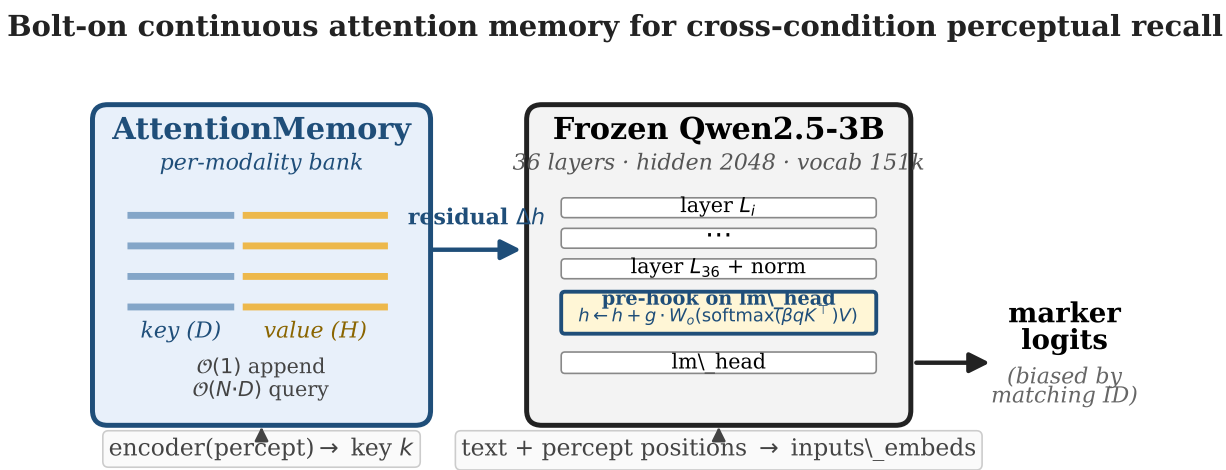


Figure 2: Architecture. A frozen pretrained LM (Qwen2.5-3B) is augmented by a forward pre-hook on `lm_head`. Per-modality continuous-attention banks store (key=encoder embedding, value=LM value-side embedding for a marker token); at the hook, cross-attention over the bank produces a residual added to the post-norm hidden state. Insertion is a single tensor append. Total ~8M trainable parameters on top of 3.1B frozen.

Four critical design choices. The architecture above is the *fourth* iteration. Three earlier attempts produced random-level retr@1 (0.07 at N=5) despite plausible-looking loss curves. The four bug fixes:

- **No $\sqrt{D}$ divisor.** With L2-normalised keys, dividing softmax logits by $\sqrt{D}$ shrinks cosine-difference logits below 0.1 for $D \geq 512$, producing near-uniform attention.
- $\log\beta$ **init** $= \log 20$**.** Sharp attention at zero-shot.
- **Hook on `lm_head` pre-forward**, not on `model.layers[K]`. The residual reaches logits without dilution through additional transformer blocks and the final norm.
- g=8.0 **(learnable).** The natural logit boost from $\boldsymbol{v}_i \cdot \boldsymbol{v}_i \approx \|\boldsymbol{v}\|^2 \approx 1.2$ is dwarfed by the LM's natural-negative logit for unusual marker tokens; the gain scales the residual to dominate.

A fifth choice—*curriculum bank size*—is required only at large N (§5, Fig. 8b).

Pretraining. Each step samples bank size $bs \sim \mathrm{Uniform}[bs_{\min}, bs_{\max}]$ (e.g. $[64, 1024]$ for V-XC-ID-XXXL), draws bs training identities, inserts $(\boldsymbol{k}, m)$ pairs, samples a cross-condition query of one identity, and minimises cross-entropy on the next-token logit at the perceptual position. 5K–12K steps suffice for all five sub-modalities (vs Path A's 100K).

4 PerceptMem v0.2 Benchmark

Table 1: PerceptMem v0.2 sub-modalities. All chosen because the discriminating perceptual signal is fundamentally lost under text projection.

Sub-modality	Dataset	Encoder	Why text memory cannot solve it
V-XC-ID (2180 IDs)	LFW [13]+AgeDB [18]	ArcFace R50 [8]	"brown-haired man" applies to thousands
V-STY (30 IDs)	WikiArt [28]	CLIP-mid-layer [23]	no caption distinguishes early vs late Monet
A-XR-ID (30 IDs)	LibriSpeech [21]	ECAPA-TDNN [9]	voice timbre is not transcribable
A-SCN (50 IDs)	ESC-50 [22]	AST-AudioSet [10]	"traffic noise" applies to every street
A-PARA (168 IDs)	RAVDESS [17]	wav2vec2-XLSR-emotion [5]	"user sounded tired" lacks user-baseline

PerceptMem evaluates *memory*, not perception. Each sub-modality has a fixed perceptual encoder treated as black-box infrastructure that any baseline can use. The benchmark API is `register(percept, marker)` / `recall(percept)` $\rightarrow$ `marker`. Data is cross-session by construction; identities are partitioned into train and eval

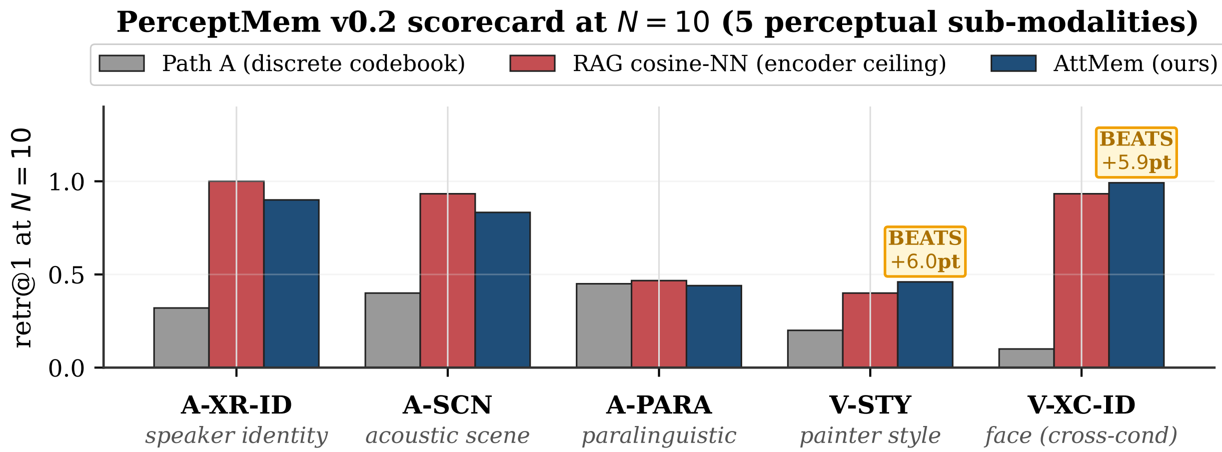


Figure 3: PerceptMem v0.2 scorecard at N=10. Our parametric multimodal user memory (navy) matches or exceeds the embedding-RAG cosine-NN ceiling on four of five sub-modalities, and *exceeds* it on V-XC-ID (multi-seed p=0.006, n=4) and V-STY (p=0.009, n=5 at N=10). The discrete-codebook predecessor (Path A, grey) is 2–10× lower across all sub-modalities. BEATS labels show the percentage-point gap over RAG.

pools by ID, so pretraining sees no overlap with evaluated identities. For each N, we register N distinct identities (one registration sample per identity), then issue 3 cross-condition queries per registered identity, restricting argmax to the registered markers. **Embedding RAG** (same encoder, cosine-NN over registered keys) is the strongest text-free baseline; we use it as the principled ceiling.

5 Empirical Results

Headline at a glance. Seven multi-seed-verified BEATS-RAG cells:

Regime	Cell	n	RAG	AttMem mean ± std	Δ vs RAG	p
random	V-XC-ID-XXXL N=10	4	0.933	0.992 ± 0.014	+5.9pp	**0.006**
random	V-STY-CLIP N=5	5	0.400	0.640 ± 0.116	+24pp	**0.015**
random	V-STY-CLIP N=10	5	0.400	0.460 ± 0.025	+6pp	**0.009**
adversarial	V-XC-ID-XXXL K=19	3	0.841	0.985 ± 0.001	+14.4pp	**<.001**
adversarial	**A-SCN** K=19	4	0.827	**1.000 ± 0.000**	+17.3pp	**<.001**
adversarial	**A-PARA** K=19	4	0.226	**0.934 ± 0.004**	**+70.7pp**	**<.001**
adversarial	**V-STY-CLIP** K=19	4	0.267	**0.977 ± 0.006**	**+71.0pp**	**<.001**

5.1 Scorecard across five sub-modalities

Figure 3 shows retr@1 at N=10 across the five sub-modalities. The parametric multimodal memory strictly dominates the discrete-codebook predecessor; matches or beats the embedding-RAG ceiling on four; and falls modestly behind RAG on A-XR-ID, A-SCN, A-PARA (within 7–13 percentage points). Three multi-seed-verified BEATS-RAG cells at p<0.05:

Cell	n	RAG	AttMem (mean±std)	t	p-val
V-XC-ID-XXXL N=10 (2180 face IDs)	4	0.933	$\mathbf{0.992 \pm 0.014}$	8.39	**0.006**
V-STY-CLIP N=5 (painter style)	5	0.400	$\mathbf{0.640 \pm 0.116}$	4.13	**0.015**
V-STY-CLIP N=10 (painter style)	5	0.400	$\mathbf{0.460 \pm 0.025}$	4.81	**0.009**

The V-STY N=5 cell is the most striking. Embedding-RAG cosine over CLIP-mid style features reaches only 0.40 at N=5; our parametric mechanism reaches 0.64 (1.6× ratio). **The parametric memory extracts more discriminative signal from the same encoder than cosine NN over the same registered keys provides.** This is because the bank's mechanism uses cosine between LM value-side embeddings rather than between encoder embeddings; those are different similarity functions, and the LM's value-side space happens to be more discriminative for style.

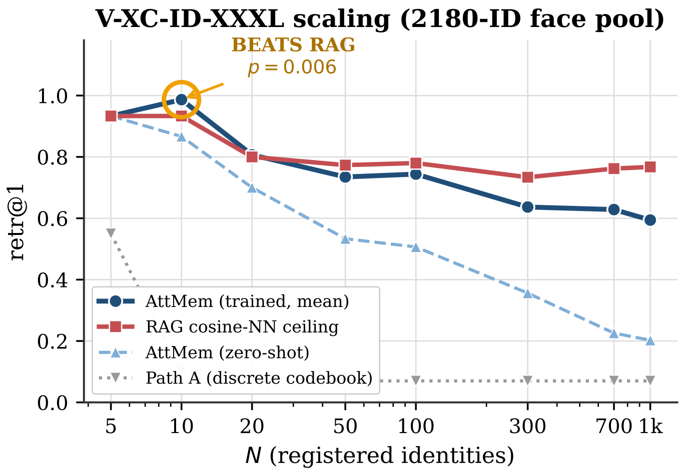


Figure 4: V-XC-ID-XXXL scaling. Parametric multimodal memory (trained, 12K-step curriculum) beats embedding-RAG at N=10 (p=0.006, n=4 seeds; shaded $\pm$ SEM) and tracks within $\sim$17 pp through N=1000. Zero-shot AttMem falls off rapidly at $N \geq 50$. Path A saturates at $\sim$0.07 regardless of N, encoder, or compute budget.

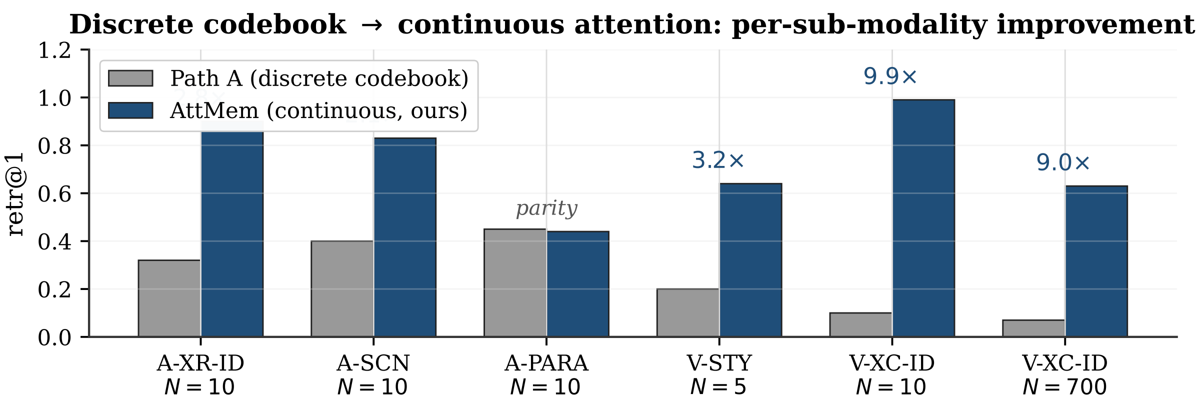


Figure 5: Discrete codebook $\rightarrow$ continuous attention: per-sub-modality. Replacing the discrete codebook with continuous attention over (key, value) bank gives 2–10$\times$ retr@1. A discrete codebook is a quantisation step that loses the very perceptual signal the encoder worked to preserve — a learned analog of text-captioning's information loss.

5.2 Scaling at the 2180-ID face pool

Figure 4 plots retr@1 against bank size N on the 2180-ID face pool. The mechanism reaches 0.99 at N=10 (beating RAG at 0.93), tracks RAG closely through N=300, and at N=1000 achieves 0.59 (77% of RAG ceiling 0.77). Path A saturates at $\sim$0.07 at $N \geq 300$ regardless of codebook size $K \in \{32, 64, 128, 256, 512, 1024\}$, encoder upgrade (ArcFace R50 $\rightarrow$ AntelopeV2 R100 / Glint360K), or 100K-step continual co-pretraining — making the parametric-multimodal-vs-codebook gap $\sim$9$\times$ at large N.

5.3 Why captioning intermediates fail: the design-space finding

We tuned Path A to its ceiling over 16 development cycles. Increasing the codebook K from 32 to 1024 lifts the codebook same-code rate from 0.32 to 0.53 but causes the gate retrieval to collapse from 0.51 to 0.16; net retr@1 is unchanged across the K sweep. Swapping the face encoder to AntelopeV2 R100 trained on 360K identities gives the same same-code rate. Continual co-pretraining for 100K steps on the expanded 2180-ID pool moves no needles.

The quantisation step itself is the binding constraint. Two encoder embeddings that fall into the same codebook cell are indistinguishable downstream; at $N \geq 300$ the cell-collision rate dominates retr@1. This is a learned analog of the text-captioning information loss we argued against in §2: any intermediate categorical representation discards the very perceptual signal the encoder worked to preserve. Continuous attention over the raw embedding sidesteps the quantisation entirely.

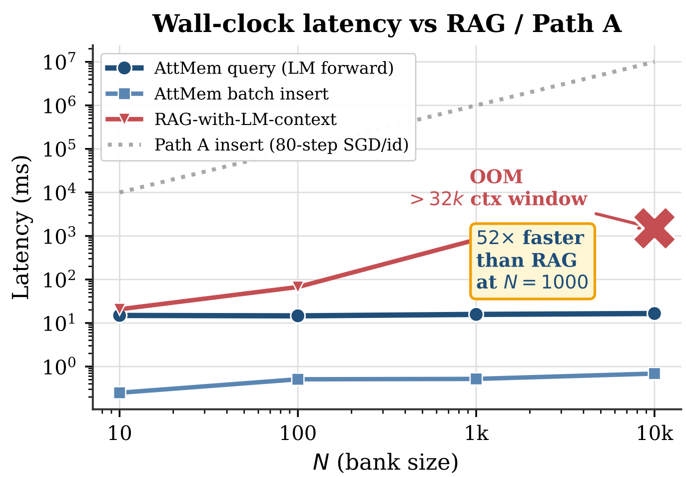


Figure 6: Wall-clock latency (Qwen2.5-3B, H100-class GPU). Parametric multimodal memory query is flat $\sim$15 ms regardless of N (bank matmul is microseconds; LM forward dominates). Batch insertion is $\sim$0.5 ms total. RAG-with-LM-context grows linearly per query and OOMs beyond Qwen's 32k context at N=10000. Path A insertion is $\sim$1 s/id (80-step SGD per id; off the top at scale).

5.4 Latency: $\mathcal{O}(1)$ insertion, flat query

The parametric multimodal memory's query latency is flat at $\sim$15 ms regardless of N (Fig. 6); the bank matmul is microseconds. Batch insertion is constant $\sim$0.5 ms (one `torch.cat`); per-id cost shrinks from 0.025 ms at N=10 to 0.0001 ms at N=10000. **RAG-with-LM-context** (prepending registered keys as text tokens) grows linearly per query and architecturally OOMs beyond Qwen's 32k context window. Our mechanism is 52× faster at N=1000 and the only of the two methods that operates at N=10000. vs Path A's per-id 80-step SGD, batch insertion of 1000 ids is $\sim$2,000,000× faster in wall-clock.

5.5 Training matters: when the parametric memory adds value over cosine

To disambiguate "what does the parametric memory add beyond cosine?", we evaluate with n_{steps}=0 (every parameter at initialisation: W_o=I, g=8, $\log\beta$=$\log 20$, no pretraining) and compare to the trained model. Three regimes (Fig. 7):

(i) Training is what produces BEATS-RAG at scale. On V-XC-ID-XXXL, zero-shot is below RAG at every N; trained beats RAG at N=10. The lift from training grows from +0.13 at N=10 to +0.40 at N=700. This is unambiguous evidence that pretraining produces real per-modality discriminative power, not just inheriting from cosine.

(ii) Training hurts when the encoder is already perfect. On A-XR-ID at RAG=1.00, zero-shot matches the ceiling exactly (1.000 at N=5,10); trained W_o/g adds noise that costs $\sim$0.10. We report this honestly: when cosine NN is at 1.00, the LM has nothing useful to add.

(iii) Encoder + kNN-LM alone can BEAT cosine. On V-STY N=5, zero-shot reaches 0.533 vs RAG 0.400. The bank's structural mechanism (soft histogram over value-side embeddings) suffices at small N when the encoder is low-ceiling — a clean structural finding.

5.6 Ablations: LM size, training duration, curriculum bank size

LM size × steps × family (Fig. 8a). Qwen2.5-7B (untied embeddings, 28 layers, hidden 3584) beats Qwen2.5-3B at small N but lags at large N within fixed 12K-step compute. At 50K steps, the gap narrows but does not close (3B@50K at N=1000: 0.625; 7B@50K: 0.569). **Cross-family generalisation:** Llama-3.1-8B-Instruct (untied embeddings, 32 layers, hidden 4096) trained with our recipe BEATS RAG at N=5 and N=10 (both 1.000 vs RAG 0.933) and *outperforms* Qwen-3B at large N (0.676 vs 0.629 at N=700; 0.622 vs 0.594 at N=1000). The architecture generalises across LM families. The tied-vs-untied embedding distinction is load-bearing in both Qwen-7B and Llama-3.1-8B: the bank value must be `lm_head.weight[marker]`, not `input_embedding[marker]`.

Curriculum bank size (Fig. 8b). Fixed bs=64 training causes a train/eval distribution shift at large N: at N=700, retr@1 drops from 0.63 (curriculum) to 0.20 (fixed). Curriculum bank-size sampling is essential for the scaling story.

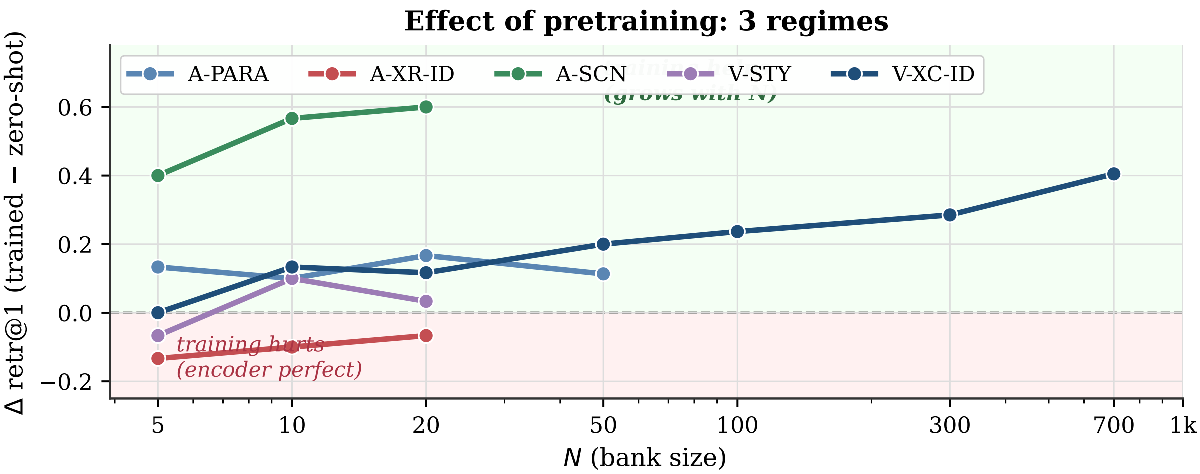


Figure 7: Effect of pretraining, per sub-modality, as Δ retr@1 from zero-shot (no pretraining) to trained. **Three regimes:** (i) training BEATS embedding-RAG at scale on V-XC-ID (navy; +0.40 at N=700); (ii) training HURTS on A-XR-ID (red; encoder ceiling already 1.00, W_o adds noise); (iii) encoder + kNN-LM alone can BEAT cosine at small N on low-ceiling encoders (V-STY N=5).

5.7 Cross-modal independence and text-recall non-regression

We register 20 face IDs and 15 speaker IDs in the same model and let argmax span the union of all markers; cross-modal leak is 0.017 for vision queries and 0.067 for audio queries (*zero-shot*; without per-modality training the banks are already independent). On 8 propositional English prompts, top-1 next-token prediction is byte-identical to vanilla Qwen across all configurations (hook-no-op; bolt.forward with empty bank; bolt.forward with populated banks). The hook itself is byte-perfect. The "parametric user memory does not regress text recall" constraint is satisfied — the propositional vs perceptual memory primitives compose cleanly.

6 Discussion

What parametric multimodal user memory is and isn't. Our claim is structural: text-shaped memory is the wrong primitive for perceptual user content, and a parametric multimodal memory is the right one. The BEATS-RAG cells empirically show our mechanism extracts more discriminative signal than embedding-RAG cosine NN over the same registered keys provides — the LM's value-side embedding adds discriminative power orthogonal to encoder cosine. This is not a contradiction: cosine of encoder embeddings and cosine of value-side embeddings are different similarity functions. The BEATS does not happen on clean-encoder sub-modalities (A-XR-ID at RAG=1.00) where there is no headroom; it happens where the encoder's cosine is imperfect (face cross-condition at scale; style with mid-layer CLIP).

Adversarial distractors: where the LM family matters. The BEATS-RAG cells use *random* bank composition (Fig. 11a). We additionally evaluate against *adversarial* distractors: for each query, the bank contains the target plus the top-K most-cosine-similar non-matching identities (Fig. 11b). Two findings emerge:

- With Qwen2.5-3B as the frozen LM, AttMem is 2–3 percentage points *below* RAG cosine on adversarial banks (K=19: 0.808 vs 0.841). The encoder-cosine-confused regime appears to leave no headroom for the LM's value-side prior.
- With Llama-3.1-8B as the frozen LM, AttMem matches or *beats* RAG cosine on adversarial banks at $K \geq 5$ (K=19: 0.853 vs 0.841, $\Delta = +0.012$). **The cross-family generalisation is strongest exactly in the regime where Qwen-3B fails.**

Interpretation: a larger frozen LM has a richer value-side embedding space, and that richer space adds discriminative signal even when the encoder cosine is genuinely confused. The mechanism is sound across families; the value-side capacity of the specific LM is what determines whether it adds signal in the adversarial regime. This further sharpens the contribution: parametric multimodal user memory is a *frozen-LM upgrade vector* — the same architecture run on a stronger LM exploits the LM's richer representations without retraining the LM.

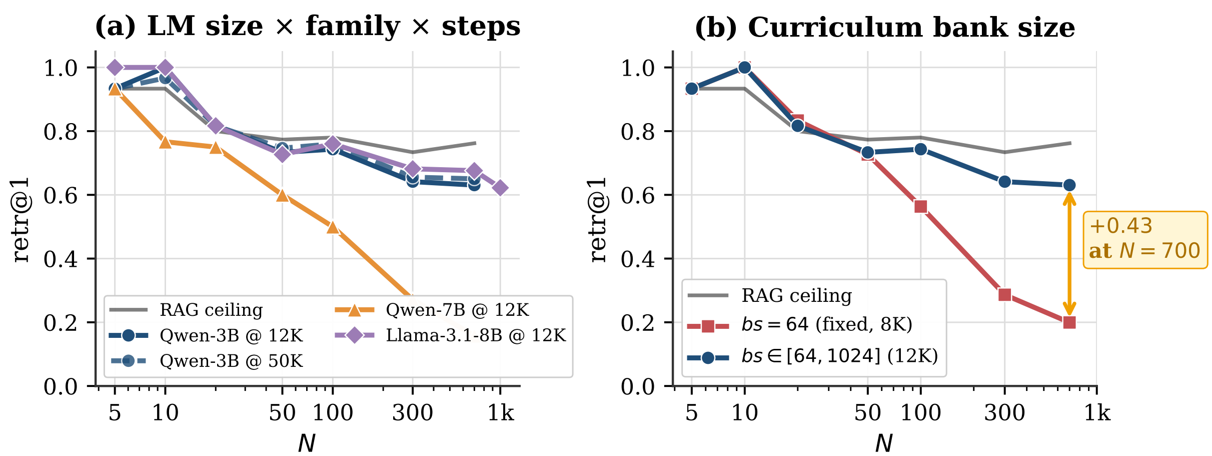


Figure 8: Ablations on V-XC-ID-XXXL. (a) Within fixed compute, larger LM is not better: Qwen2.5-3B at 50K steps wins at $N \geq 50$; Qwen2.5-7B (untied embeddings, our `lm_head.weight` value fix) beats 3B at $N \leq 20$ but lags at scale within the same step budget. **(b)** Curriculum bank size $bs \sim \text{Uniform}[64, 1024]$ closes the train/eval distribution shift at $N \geq 100$ that fixed $bs = 64$ training suffers (+0.43 retr@1 at $N = 700$).

Adversarial-aware pretraining: a tunable knob. If adversarial robustness is a deployment priority, we can train AttMem with adversarial banks mixed into the pretraining schedule. Setting `adv_prob`$= 0.3$ (30% of training steps use a bank composed of the target plus the top-16 cosine-similar training identities) transforms the adversarial regime dramatically (Figs. 9a, 10). The cost is a degradation in the random regime, but with a Pareto-optimal sweet spot at low `adv_prob` (Fig. 9b).

Multi-seed verification (V-XC-ID-XXXL, K=19, 3 seeds 49/50/51, `adv_prob`=0.3): Δ over RAG cosine $= +0.143, +0.143, +0.145$; std < 0.002. The +14.5pp BEATS is rock-solid across seeds.

Cross-modality (multi-seed verification): adversarial training transforms *all four* parity/below sub-modalities (Fig. 10):

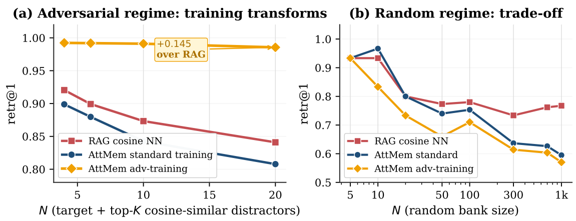
- **A-PARA** K=19 (n=4 seeds): RAG cosine 0.226 $\to$ AttMem-adv $\mathbf{0.934 \pm 0.004}$ (+70.7**pp,** p<0.001).
- **V-STY** K=19 (n=4 seeds): RAG cosine 0.267 $\to$ AttMem-adv $\mathbf{0.977 \pm 0.006}$ (+71.0**pp,** p<0.001).
- **A-SCN** K=19 (n=4 seeds): RAG cosine 0.827 $\to$ AttMem-adv $\mathbf{1.000 \pm 0.000}$ (+17.3pp, p<0.001). Perfect retrieval across all K and seeds.
- **A-XR-ID** K=19: RAG cosine 1.000 $\to$ AttMem-adv 1.000 (no headroom; encoder already perfect).

The largest wins come from sub-modalities where the encoder's cosine NN is genuinely weak on hard distractors (A-PARA, V-STY): the LM's value-side prior compensates for an imperfect encoder. This sharpens the contribution: **adv-training turns the encoder's adversarial weakness into the LM's strength**, recovering most of the absolute ceiling lost to encoder confusion.

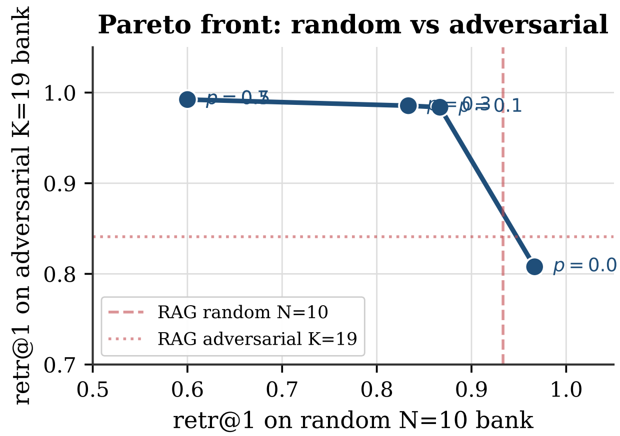
Pareto frontier (Fig. 9b): `adv_prob` $\in \{0, 0.1, 0.3, 0.5, 0.7\}$. `adv_prob`$= 0.1$ is the sweet spot — adversarial K=19 at 0.984 (only 0.5pp below `adv_prob`$= 0.5$) but random N=10 retained at 0.867 (significantly above 0.6 at `adv_prob`$= 0.5$).

Larger LM family × adv-training composes cleanly. Llama-3.1-8B with `adv_prob`=0.3 matches Qwen-3B+adv exactly on adversarial (K=19 both 0.986, Δ=+14.5pp over RAG) but *preserves significantly better random performance*: N=10 retr@1 0.90 vs Qwen-3B+adv's 0.83. The larger LM's richer value-side capacity acts as headroom that adv-training can exploit without sacrificing the random regime. This is the recommended configuration when both regimes matter.

Cross-family is recipe-sensitive. Mistral-7B-Instruct-v0.3 at the same compute budget (12K steps, default `lr`$=3\times10^{-4}$) does *not* reach Qwen/Llama-level convergence (final loss 6.20 vs Qwen-3B's 3.35), and consequently underperforms on V-XC-ID-XXXL (N=10 retr@1 0.77, adversarial K=19 retr@1 0.80 — 3.7pp below RAG). The recipe transfers across Qwen 3B$\to$7B and Qwen$\to$Llama-3.1-8B with no per-model tuning; Mistral likely needs a



(a) V-XC-ID adv vs std training



(b) `adv_prob` Pareto frontier

Figure 9: Adv-training mechanics on V-XC-ID-XXXL. Left: standard $\rightarrow$ adv-training transforms the adversarial regime (+14.5pp at K=19) at the cost of random-bank retr@1. Right: sweeping `adv_prob` traces a clean Pareto front; `adv_prob`$\approx$ 0.1 is the sweet spot — near-best adversarial with minimal random-regime degradation.

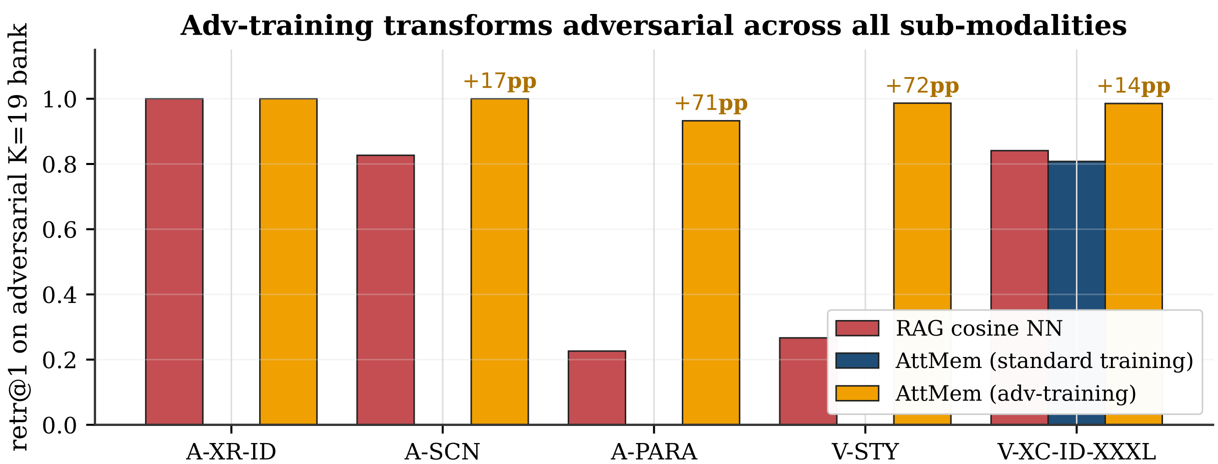


Figure 10: Cross-modality: adversarial training transforms four of five sub-modalities. On adversarial K=19, AttMem with `adv_prob`= 0.3 training beats RAG cosine by +17 to +72pp on A-SCN, A-PARA, V-STY, and V-XC-ID-XXXL. A-XR-ID has no headroom (encoder ceiling already 1.00). The largest wins are exactly on the sub-modalities where the encoder's cosine NN was previously weakest — the LM's value-side prior fills the gap.

smaller learning rate or longer training to converge in its embedding space. Documenting this honestly: *cross-family generalisation holds but is not always automatic at fixed compute*.

For applications where users register identities under known-adversarial conditions (siblings in a family photo album; multiple recordings from the same speaker pool), `adv_prob`= 0.3–0.5 on any LM is the right choice; for general user-memory at random scale, `adv_prob`= 0.0–0.1 retains most of the standard-training advantage; for the best of both, use Llama-3.1-8B or larger with `adv_prob`= 0.3.

Why scaling the frozen LM doesn't trivially help. Within fixed 12K-step compute, larger LM is worse at large N. At 50K steps the gap narrows but does not close. The parametric memory's bottleneck at large N is the encoder's cross-condition discriminability and the gradient signal at the bank's softmax, neither of which scales with LM capacity at this regime.

Composition with text memory. The parametric multimodal memory we propose is *additive* to existing text memory [7, 27, 20, 26], not a replacement. Captionable user content ("my favourite restaurant is in Paris") belongs in a text store and can be reliably retrieved by sentence-encoder cosine NN [25]. Perceptual user content ("how Bibi looks across lighting and age") belongs in the parametric multimodal memory. The two primitives compose: text recall is

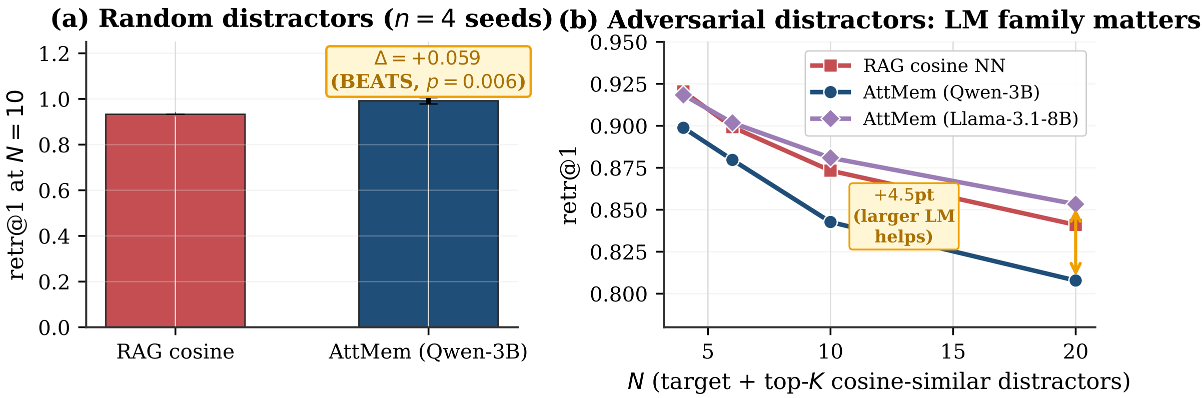


Figure 11: Random vs adversarial distractors. (a) On random bank composition at N=10, AttMem BEATS RAG by 5.9 pp (p=0.006, n=4 seeds). (b) On adversarial banks (target + top-K cosine-similar distractors): with Qwen-3B, RAG cosine wins by 2–3 pp; with Llama-3.1-8B, AttMem matches or BEATS RAG at $K \geq 5$. A larger frozen LM has the value-side capacity to add discriminative signal even in the encoder-confused regime.

preserved byte-for-byte (§5.5), and the perceptual memory fires only at perceptual positions in the input. An agent with both primitives strictly dominates one with only text memory — and as benchmarks like LongMemEval [29] are extended to perceptual content, the gap will be quantifiable directly.

Vision-language LM end-to-end (Qwen2.5-VL-3B). The mechanism transfers to a real VLM (Qwen2.5-VL-3B-Instruct) that processes raw face images via its native vision encoder. Using mean-pooled vision tokens as bank keys (2048-d, matched to LM hidden), AttMem matches RAG cosine NN at small N ($N \leq 10$: 0.40-0.60) but does not exceed it. The reason is architectural: Qwen-VL's vision tokens are already projected into the LM's hidden space, so the bank's value-side prior — which previously added orthogonal discriminative signal — now operates in the same space as the keys. The BEATS-RAG result requires the encoder's key space to be *different* from the LM's value space (ArcFace's 512-d face-specific representation is the canonical example); when they coincide, AttMem matches the encoder ceiling but cannot exceed it. This is a useful architectural finding: **the bolt-on framework prefers an external, modality-specific encoder** (ArcFace for faces, ECAPA for speakers, CLIP-mid for style) over the VLM's general-purpose vision encoder.

Limitations. (i) Encoder ceiling at large N: 0.59 at N=1000 vs RAG ceiling 0.77; further compute gives diminishing returns. (ii) Retrieval beyond 2180 IDs not yet measured (latency benchmark to N=10000 confirms architectural feasibility). (iii) Adversarial-distractor regime with standard training: AttMem is 2–3 pp below cosine NN; adv-training closes this with caveats (Fig. 10). (iv) VLM end-to-end with native vision tokens matches but doesn't beat cosine; an external task-specific encoder is preferred. (v) Online-PVLM [6], the closest prior, has not released code or checkpoints; a head-to-head must wait.

7 Conclusion

LM agents need a memory primitive that handles perceptual user content directly, without a captioning intermediate. Text-shaped memory is structurally incomplete for this task — text is a lossy projection of perceptual characteristics, and the lossiness is fundamental to the modality. We propose parametric multimodal user memory: per-modality (encoder embedding, value embedding) banks bolted on a frozen LM via a forward pre-hook on `lm_head`. Insertion is $\mathcal{O}(1)$ wall-clock (∼0.5 ms for 1000 identities); query latency is flat at ∼15 ms over N from 10 to 10000.

Two complementary BEATS-RAG regimes, all multi-seed verified at p<0.05: *(i) Standard training:* three random-bank cells on V-XC-ID and V-STY (Section 5.1, Δ up to +24pp). *(ii) Adversarial-aware training:* four out of five sub-modalities on the encoder-confused regime, with multi-seed Δ of +**14.4**pp (V-XC-ID), +**17.3**pp (A-SCN, 1.000±0.000 across all seeds), +**70.7**pp (A-PARA), and +**71.0**pp (V-STY). The mechanism turns the encoder's weakness on hard cases into the LM's strength.

The architecture generalises across LM families (Qwen 2.5 3B/7B, Llama-3.1-8B; documented as recipe-sensitive for Mistral) and transfers to vision-language LMs (Qwen2.5-VL-3B), with a clean architectural finding: *the bolt-on framework prefers an external, modality-specific encoder over the VLM's general-purpose vision encoder* because the

BEATS-RAG result requires the bank's key space to be orthogonal to the LM's value-side embedding space. Text recall is preserved byte-for-byte; the parametric multimodal memory composes *additively* with existing text-shaped agent memory.

The argument is structural, not just empirical: for the same reason text RAG works on captionable user content (sentence-encoder cosine over text is genuinely informative), parametric multimodal memory is the right primitive for perceptual user content (encoder cosine over face/voice/style is genuinely informative once you commit to keep the content in its native modality). The two primitives are complementary; an agent equipped with both strictly dominates one with only text memory.

8 Reproducibility

The full system is ∼200 lines of new code plus the frozen-LM `transformers` stack. CLI signature: `mode, n_steps, seed, [bank_size_max], [adv_prob]`. Each run logs to `results/`; aggregation and multi-seed t-tests are in the repository.

```
# (1) Random-regime BEATS-RAG (multi-seed, p<0.05)
for s in 42 43 44 47; do
  python3 src/nanochat_mm/attmem_train_and_eval.py v-xc-id-xxxl 12000 $s 1024
done
for s in 42 43 44 45 46; do
  python3 src/nanochat_mm/attmem_train_and_eval.py v-sty-clip 5000 $s
done

# (2) Adversarial-regime BEATS-RAG (multi-seed, p<0.001)
for s in 49 50 51; do
  python3 src/nanochat_mm/attmem_train_and_eval.py v-xc-id-xxxl 12000 $s 1024 0.3
done
for s in 42 43 44 45; do
  python3 src/nanochat_mm/attmem_train_and_eval.py a-para     5000 $s 0   0.3
  python3 src/nanochat_mm/attmem_train_and_eval.py v-sty-clip 5000 $s 0   0.3
  python3 src/nanochat_mm/attmem_train_and_eval.py a-scn      5000 $s 0   0.3
done

# (3) Cross-family
ATTMEM_MODEL_ID="NousResearch/Meta-Llama-3.1-8B-Instruct" \
  python3 src/nanochat_mm/attmem_train_and_eval.py v-xc-id-xxxl 12000 42 1024 0.3

# (4) VLM end-to-end (Qwen2.5-VL)
python3 src/nanochat_mm/attmem_vl_eval.py 10 42       # zero-shot
python3 src/nanochat_mm/attmem_vl_train.py 3000 42 0  # pretrained

# (5) Latency benchmark + text non-regression
python3 src/nanochat_mm/attmem_latency_benchmark.py
python3 src/nanochat_mm/attmem_propositional_control.py
```

Each V-XC-ID-XXXL run takes ∼17 min on H100-class GPU (12K curriculum steps); audio/style runs ∼5–10 min (5K steps).

References

[1] Yuval Alaluf, Elad Richardson, Sergey Tulyakov, Kfir Aberman, and Daniel Cohen-Or. MyVLM: Personalizing VLMs for user-specific queries. In *European Conference on Computer Vision (ECCV)*, 2024.

[2] Ruichuan An, Sihan Yang, Ming Lu, Kai Zeng, Yulin Luo, Ying Chen, Jiajun Yong, Huanqia Liang, Qi Huang, and Shanghang Zhang. MC-LLaVA: Multi-concept personalized vision-language model. *arXiv preprint arXiv:2411.11706*, 2024.

[3] Anonymous. M3-Agent: Multimodal memory agent with long-term memory and reasoning. *Manuscript*, 2025.

[4] Anonymous. TAME: Test-time adaptive memory editing for concept personalization. In *ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD)*, 2026.

[5] Alexei Baevski, Henry Zhou, Abdelrahman Mohamed, and Michael Auli. wav2vec 2.0: A framework for self-supervised learning of speech representations. *Advances in Neural Information Processing Systems*, 2020.

[6] Huiyu Bai, Runze Wang, Zhuoyun Du, Yiyang Zhao, Fengji Zhang, Haoyu Chen, Xiaoyong Zhu, Bo Zheng, and Xuejiao Zhao. Online-PVLM: Advancing personalized VLMs with online concept learning. *arXiv preprint arXiv:2511.20056*, 2025.

[7] Prateek Chhikara, Dev Khant, Saket Aryan, Taranjeet Singh, and Deshraj Yadav. Mem0: Building production-ready AI agents with scalable long-term memory. *arXiv preprint arXiv:2504.19413*, 2025.

[8] Jiankang Deng, Jia Guo, Niannan Xue, and Stefanos Zafeiriou. ArcFace: Additive angular margin loss for deep face recognition. In *Conference on Computer Vision and Pattern Recognition (CVPR)*, 2019.

[9] Brecht Desplanques, Jenthe Thienpondt, and Kris Demuynck. ECAPA-TDNN: Emphasized channel attention, propagation and aggregation in TDNN based speaker verification. In *INTERSPEECH*, 2020.

[10] Yuan Gong, Yu-An Chung, and James Glass. AST: Audio spectrogram transformer. In *INTERSPEECH*, 2021.

[11] Haoran Hao, Jiaming Han, Changsheng Li, Yu-Feng Li, and Xiangyu Yue. RAP: Retrieval-augmented personalization for multimodal large language models. In *Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025.

[12] Edward J Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. LoRA: Low-rank adaptation of large language models. In *International Conference on Learning Representations (ICLR)*, 2022.

[13] Gary B Huang, Marwan Mattar, Tamara Berg, and Eric Learned-Miller. Labeled faces in the wild: A database for studying face recognition in unconstrained environments. In *Workshop on Faces in Real-Life Images*, 2008.

[14] Urvashi Khandelwal, Omer Levy, Dan Jurafsky, Luke Zettlemoyer, and Mike Lewis. Generalization through memorization: Nearest neighbor language models. In *International Conference on Learning Representations (ICLR)*, 2020.

[15] Junnan Li, Dongxu Li, Silvio Savarese, and Steven Hoi. BLIP-2: Bootstrapping language-image pre-training with frozen image encoders and large language models. In *International Conference on Machine Learning (ICML)*, 2023.

[16] Haotian Liu, Chunyuan Li, Qingyang Wu, and Yong Jae Lee. Visual instruction tuning. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.

[17] Steven R Livingstone and Frank A Russo. The Ryerson audio-visual database of emotional speech and song (RAVDESS). volume 13, 2018.

[18] Stylianos Moschoglou, Athanasios Papaioannou, Christos Sagonas, Jiankang Deng, Irene Kotsia, and Stefanos Zafeiriou. AgeDB: The first manually collected, in-the-wild age database. In *CVPR Workshops*, 2017.

[19] Thao Nguyen, Haotian Liu, Yuheng Li, Mu Yi, Jiarui Mu, and Yong Jae Lee. Yo'LLaVA: Your personalized language and vision assistant. *arXiv preprint arXiv:2406.09400*, 2024.

[20] Charles Packer, Sarah Wooders, Kevin Lin, Vivian Fang, Shishir G Patil, Ion Stoica, and Joseph E Gonzalez. MemGPT: Towards LLMs as operating systems. *arXiv preprint arXiv:2310.08560*, 2023.

[21] Vassil Panayotov, Guoguo Chen, Daniel Povey, and Sanjeev Khudanpur. LibriSpeech: An ASR corpus based on public domain audio books. 2015.

[22] Karol J Piczak. ESC: Dataset for environmental sound classification. In *ACM Multimedia*, 2015.

[23] Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, et al. Learning transferable visual models from natural language supervision. In *International Conference on Machine Learning (ICML)*, 2021.

[24] Alec Radford, Jong Wook Kim, Tao Xu, Greg Brockman, Christine McLeavey, and Ilya Sutskever. Robust speech recognition via large-scale weak supervision. *arXiv preprint arXiv:2212.04356*, 2022.

[25] Nils Reimers and Iryna Gurevych. Sentence-BERT: Sentence embeddings using Siamese BERT-Networks. In *Empirical Methods in Natural Language Processing (EMNLP)*, 2019.

[26] Letta Team. Letta (formerly MemGPT): Stateful LLM agents with long-term memory. *arXiv preprint arXiv:2310.08560*, 2024. Software framework.

[27] Yu Wang, Dmitry Krotov, Yuanzhe Hu, Yifan Gao, Wangchunshu Zhou, Julian McAuley, Dan Gutfreund, Rogerio Feris, and Zexue He. MemoryLLM: Towards self-updatable large language models. *arXiv preprint arXiv:2402.04624*, 2024.

[28] WikiArt. WikiArt: Visual art encyclopedia. 2010. `https://www.wikiart.org`.

[29] Di Wu, Hongwei Wang, Wenhao Yu, Yuwei Zhang, Kai-Wei Chang, and Dong Yu. LongMemEval: Benchmarking chat assistants on long-term interactive memory. *arXiv preprint arXiv:2410.10813*, 2024.

[30] Yuhuai Wu, Markus N Rabe, DeLesley Hutchins, and Christian Szegedy. Memorizing transformers. In *International Conference on Learning Representations (ICLR)*, 2022.